$$\frac{}{(\textbf{skip};,|\psi\rangle,A,l)\xrightarrow{0}(\top,|\psi\rangle,A,l)} \qquad \frac{(\mathrm{q},l)\Downarrow_{\mathbb{N}} n\notin A}{(\mathrm{q}\ \texttt{*=}\ \mathrm{U}^f(\mathrm{j});,|\psi\rangle,A,l)\xrightarrow{0}(\bot,|\psi\rangle,A,l)}$$

$$\frac{(\mathrm{q},l)\Downarrow_{\mathbb{N}} n\in A \qquad (\mathrm{j},l)\Downarrow_{\mathbb{Z}} k}{(\mathrm{q}\ \texttt{*=}\ \mathrm{U}^f(\mathrm{j});,|\psi\rangle,A,l)\xrightarrow{0}(\top,I_{2^{n-1}}\otimes[\![\mathrm{U}]\!](f)(k)\otimes I_{2^{l(|\psi\rangle)-n}}|\psi\rangle,A,l)}$$

$$\frac{(\mathrm{S}_1,|\psi\rangle,A,l)\xrightarrow{m_1}(\top,|\psi'\rangle,A,l) \qquad (\mathrm{S}_2,|\psi'\rangle,A,l)\xrightarrow{m_2}(\diamond,|\psi''\rangle,A,l)}{(\mathrm{S}_1\ \mathrm{S}_2,|\psi\rangle,A,l)\xrightarrow{m_1+m_2}(\diamond,|\psi''\rangle,A,l)}$$

$$\frac{(\mathrm{S}_1,|\psi\rangle,A,l)\xrightarrow{m}(\bot,|\psi\rangle,A,l)}{(\mathrm{S}_1\ \mathrm{S}_2,|\psi\rangle,A,l)\xrightarrow{m}(\bot,|\psi\rangle,A,l)} \qquad \frac{(\mathrm{b},l)\Downarrow_{\mathbb{B}} b \qquad (\mathrm{S}_b,|\psi\rangle,A,l)\xrightarrow{m_b}(\diamond,|\psi'\rangle,A,l) \qquad \diamond\in\{\top,\bot\}}{(\textbf{if}\ \mathrm{b}\ \textbf{then}\ \mathrm{S}_{\textbf{true}}\ \textbf{else}\ \mathrm{S}_{\textbf{false}},|\psi\rangle,A,l)\xrightarrow{m_b}(\diamond,|\psi'\rangle,A,l)}$$

$$\frac{(\mathrm{q},l)\Downarrow_{\mathbb{N}} n\in A \qquad \forall k\in\{0,1\},\ (\mathrm{S}_k,|\psi\rangle,A\backslash\{n\},l)\xrightarrow{m_k}(\top,|\psi_k\rangle,A\backslash\{n\},l)}{(\textbf{qcase}\ \mathrm{q}\ \textbf{of}\ \{0\to\mathrm{S}_0,1\to\mathrm{S}_1\},|\psi\rangle,A,l)\xrightarrow{\max_{k\in\{0,1\}}m_k}(\top,\sum_k|k\rangle_n\langle k|_n|\psi_k\rangle,A,l)}$$

$$\frac{(\mathrm{q},l)\Downarrow_{\mathbb{N}} n\in A \qquad \forall k\in\{0,1\},\ (\mathrm{S}_k,|\psi\rangle,A\backslash\{n\},l)\xrightarrow{m_k}(\diamond_k,|\psi_k\rangle,A\backslash\{n\},l) \qquad \bot\in\{\diamond_0,\diamond_1\}}{(\textbf{qcase}\ \mathrm{q}\ \textbf{of}\ \{0\to\mathrm{S}_0,1\to\mathrm{S}_1\},|\psi\rangle,A,l)\xrightarrow{\max_{k\in\{0,1\}}m_k}(\bot,|\psi\rangle,A,l)}$$

$$\frac{(\mathrm{q},l)\Downarrow_{\mathbb{N}} n\notin A}{(\textbf{qcase}\ \mathrm{q}\ \textbf{of}\ \{0\to\mathrm{S}_0,1\to\mathrm{S}_1\},|\psi\rangle,A,l)\xrightarrow{0}(\bot,|\psi\rangle,A,l)} \qquad \frac{(\mathrm{s},l)\Downarrow_{\mathcal{L}(\mathbb{N})}[\,]}{(\textbf{call}\ \texttt{proc}(\mathrm{s});,|\psi\rangle,A,l)\xrightarrow{1}(\top,|\psi\rangle,A,l)}$$

$$\frac{(\mathrm{s},l)\Downarrow_{\mathcal{L}(\mathbb{N})} l'\neq[\,] \qquad (\mathrm{S}^{\texttt{proc}},|\psi\rangle,A,l')\xrightarrow{m}(\diamond,|\psi'\rangle,A,l') \qquad \diamond\in\{\top,\bot\}}{(\textbf{call}\ \texttt{proc}(\mathrm{s});,|\psi\rangle,A,l)\xrightarrow{m+1}(\diamond,|\psi'\rangle,A,l)}$$